CLINICAL · APPROACH TO

# Approach to stridor and the threatened airway

Stridor is a sign, not a diagnosis: the sound of air forced through a narrowed airway. Its timing tells you where the narrowing is, its history tells you what it is, and the work of breathing tells you how long you have. Judge the severity first, keep the patient calm and upright, and never let a full examination or a scan come before a safe airway.

**AT A GLANCE**

- Noise timing tells the level.
- Keep calm, sit up, call early.
- Use adrenaline and steroids as a bridge.
- Have a surgical airway ready.

## 1 CHAPTER ONE Definition

Noise and where it comes from.

### 1.1 Stridor

A harsh, noisy breathing sound produced by turbulent airflow through a narrowed larynx or trachea. *Stertor* is the lower-pitched snoring of the nose, nasopharynx or oropharynx; *wheeze* is the musical expiratory sound of the smaller airways.

### 1.2 Phase and site

- **Inspiratory:** supraglottis, pharynx and extrathoracic airway collapse on inspiration (laryngomalacia, epiglottitis, retropharyngeal abscess, supraglottic tumours).
- **Biphasic:** a fixed narrowing at the glottis, subglottis or cervical trachea (subglottic stenosis, bilateral cord palsy, papillomas, croup).
- **Expiratory:** the intrathoracic trachea and bronchi (tracheomalacia, bronchial foreign body, tracheal stenosis, compression by masses or vessels).

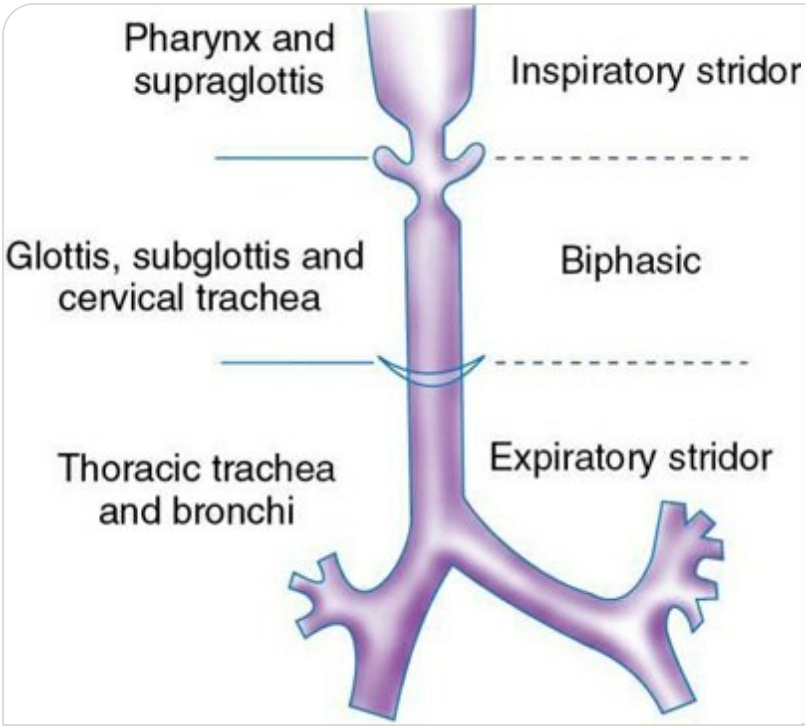



**FIGURE 1** Types of stridor and their site of origin.
Dhingra, fig. 59.3

## 1.3 Associated sounds

- Snoring or snorting: nose and nasopharynx.
- Gurgling and a muffled voice: pharynx and supraglottis.
- Hoarse voice or weak cry: the vocal folds. The cry is normal in laryngomalacia and subglottic stenosis.
- Barking cough: subglottis (croup).
- Expiratory wheeze: bronchi.

# 2 CHAPTER TWO Causes

From the nose to the bronchi.

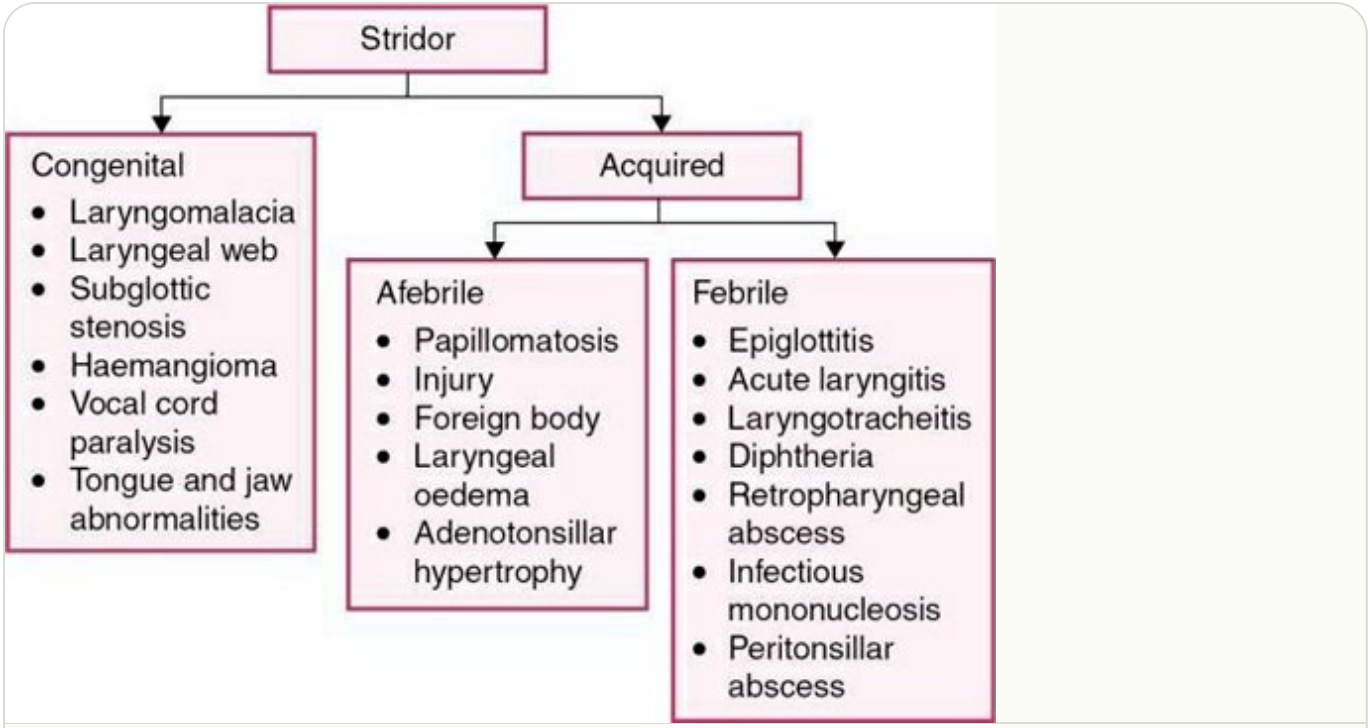



**FIGURE 2** Causes of stridor in infants and children: congenital, and acquired with or without fever. Dhingra, fig. 59.5

## CAUSES OF STRIDOR BY SITE

| SITE | CONGENITAL | ACQUIRED |
|---|---|---|
| Nose | Choanal atresia | Severe nasal obstruction in neonates |
| Tongue and jaw | Macroglossia, lingual thyroid, micrognathia (Pierre Robin) | Haemangioma, lymphangioma, Ludwig's angina |
| Pharynx | Dermoid, cysts | Adenotonsillar hypertrophy, retropharyngeal abscess, tumours |
| Larynx | Laryngomalacia, web, cysts, vocal fold paralysis, subglottic stenosis, haemangioma, cleft | Epiglottitis, croup, diphtheria, oedema, foreign body, trauma, papillomas, carcinoma, bilateral palsy, post-intubation stenosis, tetany, laryngospasm |

| SITE | CONGENITAL | ACQUIRED |
|---|---|---|
| Trachea and bronchi | Tracheomalacia, stenosis, atresia | Bacterial tracheitis, foreign body, stenosis after intubation or tracheostomy, tumours |
| Outside the airway | Vascular rings, oesophageal atresia and fistula, congenital goitre, cystic hygroma | Oesophageal foreign body, goitre, mediastinal and neck masses |

### 2.1 In adults

Laryngeal and tracheal tumours, bilateral vocal fold paralysis (after thyroidectomy), post-intubation and tracheostomy stenosis, laryngeal trauma, angioedema, epiglottitis, deep neck infection, goitre and foreign bodies.

## 3 CHAPTER THREE Assessment

How bad, and what.

### 3.1 Severity

- **Mild:** stridor only when active or crying, no recession.
- **Moderate:** stridor at rest, suprasternal, intercostal, subcostal and sternal recession, tracheal tug, nasal flaring, tachypnoea, tachycardia.
- **Severe:** marked recession, agitation and restlessness (hypoxia), drooling, inability to speak or feed.
- **Impending arrest:** fatigue, falling respiratory rate and quieter stridor (less air moving, not improvement), pallor or cyanosis, drowsiness.



**FIGURE 3** A child with bilateral abductor paralysis: recession of the suprasternal notch, sternum, intercostal spaces and epigastrium on inspiration.

Dhingra, fig. 59.4

**A QUIETER CHILD IS NOT ALWAYS A BETTER CHILD**

Diminishing stridor with tiring, drowsiness or a falling oxygen saturation means respiratory failure. Oximetry can stay normal until late; watch the work of breathing and the level of consciousness.

## 3.2 History

- **Onset:** at birth or soon after (congenital), or later (acquired).
- **Speed:** sudden (foreign body, oedema, epiglottitis) or gradual and progressive (laryngomalacia, haemangioma, papillomas, stenosis, tumours).
- **Duration and course:** short (infection, foreign body) or long-standing.
- **Fever and toxicity:** epiglottitis, tracheitis, retropharyngeal abscess, diphtheria.
- **Feeding:** choking, cyanosis and aspiration (cleft, paralysis, fistula, vascular ring, oesophageal foreign body).
- **Cyanotic spells** and apnoeas: need for airway support.
- **Witnessed choking** on a toy, nut or food.
- **Previous intubation,** surgery (thyroid, cardiac) or trauma.

- **Voice and cry** quality.

## 3.3 Examination

Observe from a distance first: posture, colour, recession, drooling, voice. Note the phase of stridor, the effect of position (prone improves laryngomalacia, micrognathia and macroglossia), and auscultate over the nose, open mouth, neck and chest to localise. Examine the nose, tongue, jaw and pharynx; in adults, and in calm older children, flexible nasendoscopy shows the larynx.

**FLEXIBLE ENDOSCOPY IN THE CLINIC**

Under topical anaesthesia, the awake child can be examined for laryngomalacia, vocal fold movement, cysts, papillomas and webs, which an anaesthetised child cannot show: dynamic problems need a breathing patient.

## 3.4 Investigations

Chosen by the history, and only when the airway allows:

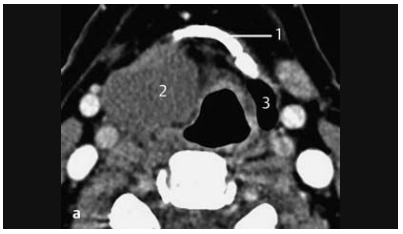
- Lateral and anteroposterior soft tissue radiographs of the neck (steeple sign of croup, thumb sign of epiglottitis, prevertebral swelling, radio-opaque foreign bodies).
- Chest radiographs, with inspiratory and expiratory films or fluoroscopy for radiolucent bronchial foreign bodies (air trapping, mediastinal shift).
- CT or MRI for subglottic stenosis, haemangioma, masses and vascular rings; angiography for vascular anomalies.
- Contrast swallow for clefts, fistulae, rings and aspiration.
- Arterial blood gases in severe obstruction.

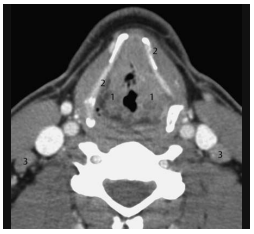
## 3.5 Microlaryngoscopy and bronchoscopy

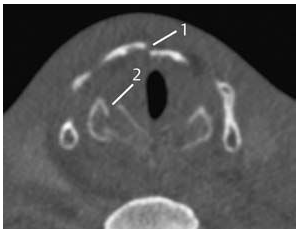
The definitive assessment of the paediatric airway, in theatre with full resuscitation facilities and an experienced anaesthetist. Inhalational induction maintains spontaneous breathing; a quick laryngoscopy is followed by bronchoscopy of the subglottis, trachea and bronchi, collection of secretions and removal of crusts or foreign bodies; then the child is intubated if needed. Look at the whole airway: a second lesion is common.

**IMAGING IN THE APP** 3 cases

Laryngocele, supraglottic tumour and laryngeal trauma on imaging.






**Laryngocele**
Infection & inflammation · 10 images

**Supraglottic Tumor**
Tumours · 9 images

**Laryngeal Trauma**
Trauma · 10 images

## 3.6 The common paediatric causes

**Laryngomalacia** is the commonest cause of stridor in infants, accounting for well over half of congenital cases. Inspiratory stridor begins in the first weeks, is worse with feeding, crying and lying supine, and usually resolves by 18 to 24 months. Short aryepiglottic folds, an omega-shaped epiglottis and prolapsing arytenoid mucosa are seen on awake flexible laryngoscopy. Most infants need only reassurance and treatment of reflux; poor weight gain, apnoeas, cyanosis or pectus excavatum are indications for **supraglottoplasty**.

**Vocal fold paralysis** is the second commonest congenital cause; bilateral paralysis is associated with Arnold–Chiari malformation and hydrocephalus. **Subglottic stenosis** is congenital or, more often, acquired after intubation, and is graded by the **Cotton–Myer** system by the percentage of lumen obstructed: grade I up to 50%, grade II 51 to 70%, grade III 71 to 99%, grade IV no lumen. **Subglottic haemangioma** typically presents at a few weeks of age with biphasic stridor that worsens over the first months, often with a cutaneous haemangioma in a beard distribution, and responds to **propranolol**. **Croup** (viral laryngotracheobronchitis) produces a barking cough and stridor in children aged six months to three years and is treated with a single dose of oral dexamethasone, 0.15 mg/kg, with nebulised adrenaline for severe distress. Recurrent or atypical croup, or croup under six months, warrants airway endoscopy.

# 4 CHAPTER FOUR Management

Airway first.

## 4.1 Immediate

1. Keep the patient calm and upright (in a parent's arms for a child); give humidified oxygen without causing distress.
2. Call senior anaesthetic and ENT help early.
3. Temporising measures as appropriate: nebulised adrenaline, steroids (dexamethasone), heliox (a helium–oxygen mixture flows more smoothly through narrow airways).
4. Treat the cause where possible: antibiotics, removal of a foreign body.

## 4.2 Securing the airway

- **Intubation:** in theatre, by the most experienced anaesthetist, with an ENT surgeon scrubbed and ready.
- **Tracheostomy:** when intubation is impossible or unsafe, for prolonged obstruction, laryngeal trauma, and for long-term airway management (bilateral paralysis, stenosis, severe congenital anomalies).
- **Cricothyroidotomy:** the emergency surgical airway in adults and older children when the patient can neither be intubated nor oxygenated; needle cricothyroidotomy with jet ventilation in young children.

## 4.3 The adult threatened airway

The Difficult Airway Society guidelines end with **front-of-neck access** when the patient "can't intubate, can't oxygenate": the recommended technique is a **scalpel–bougie–tube** cricothyroidotomy through the cricothyroid membrane, identified beforehand by palpation or ultrasound. In a patient with a known obstructing laryngeal tumour, the safest plan is often an **awake tracheostomy under local anaesthesia**, or awake fibreoptic intubation, agreed between surgeon and anaesthetist before any sedation. **High-flow humidified nasal oxygen** extends the

time before desaturation during induction and airway surgery. Heliox and steroids buy time but do not treat the cause.

## 5 CHAPTER FIVE Pathway

Assess, stabilise, secure.

### THE CHILD OR ADULT WITH STRIDOR

**1 How severe?**

- → **silent chest, cyanosis, exhaustion or falling consciousness**
  emergency airway now — theatre, senior anaesthetist and ENT
- → **severe recession or agitation**
  oxygen, nebulised adrenaline, dexamethasone, senior help
- → **mild**
  step 2

**2 Phase and age**

- → **inspiratory in an infant, worse feeding and supine**
  laryngomalacia — awake flexible laryngoscopy
- → **biphasic, worsening in first months**
  subglottic haemangioma or stenosis
- → **barking cough, toddler**
  croup — dexamethasone
- → **sudden onset while eating or playing**
  foreign body — bronchoscopy

### 3 Febrile and toxic?

- **drooling, sitting forward**
  supraglottitis — airway in theatre
- **high fever, poor response to croup treatment**
  bacterial tracheitis

### 4 Adult

- **tumour**
  awake tracheostomy or planned intubation, then biopsy
- **post-thyroidectomy bilateral palsy**
  reintubate or tracheostomy
- **angioedema**
  adrenaline, steroids, antihistamine
- stop ACE inhibitor

**OPERATIVE GUIDES IN THEATRE** 6 operations

Tracheostomy, laryngomalacia and subglottic stenosis surgery, airway foreign bodies and acquired stenosis.

**Open Tracheostomy**
Tracheostomy
**Percutaneous Tracheostomy**
Percutaneous
**Laryngomalacia**
Laryngomalacia
**Subglottic Stenosis**
Subglottic stenosis
**Tracheoscopy, Bronchoscopy, and Airway Foreign Bodies**
Airway foreign body
**Acquired Laryngeal Stenosis**
Acquired stenosis

CHAPTER SIX

# 6 For the examination

What the examiner wants to hear.

## 6.1 Viva essentials

**WHAT THE EXAMINER IS LOOKING FOR**

- **Laryngomalacia:** features, natural history and indications for supraglottoplasty.
- **Cotton–Myer grading** and subglottic haemangioma with propranolol.
- **Front-of-neck access** and the awake tracheostomy.
- **Definition** of stridor, and stertor and wheeze.
- **Inspiratory, biphasic, expiratory** and their sites.
- **Causes** in children by site, congenital and acquired.
- **History points** and severity signs.
- **Management:** airway first, then diagnosis by endoscopy.

## 6.2 Common pitfalls

**COMMON PITFALLS**

- Sending a child with stridor for radiographs unaccompanied.
- Mistaking quieter stridor in a tiring child for improvement.
- Stopping at the first lesion found and missing a second.

Written for this app from Oxford Textbook of Otolaryngology ch. 52, 74, Oxford Specialist Handbook ch. 3, 23, Oxford Handbook of ENT ch. 3, 25, Dhingra ch. 59, 64 and K. J. Lee ch. 50, 53. Figures reproduced from the books for private study. Study reference only: follow local protocols and senior advice for patients.

IMAGING

# On imaging

*The imaging of this topic: key findings, differentials and representative scans.*

LARYNX & TRACHEA · INFECTION & INFLAMMATION

## Laryngocele

### IMAGING FINDINGS

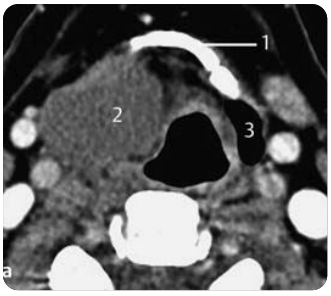
- Air or fluid-filled sac from the laryngeal ventricle
- Internal (paraglottic) or external through the thyrohyoid membrane
- Rim enhancement if infected (pyolaryngocele)

### TOP DIFFERENTIALS

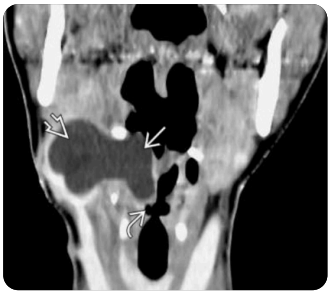
- Thyroglossal duct cyst
- Second branchial cleft cyst
- Saccular cyst

### CHECKLIST

- Exclude a carcinoma at the ventricle in adults
- Airway compromise




CT with intravenous contrast enhancement, axial. The axial slice at the lower border of the hyoid (1) reveals a homogeneous, expanding…



Coronal CECT reformat shows a similar lesion. This is also known as a "mixed" laryngocele because it contains internal…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 292.

LARYNX & TRACHEA · TUMOURS

## Supraglottic Tumor

### IMAGING FINDINGS

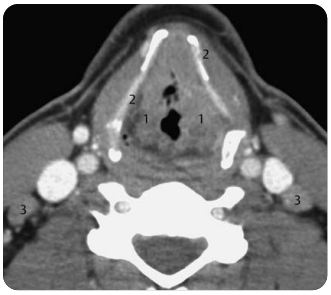
- Mass on the epiglottis, aryepiglottic folds or false cords
- Invasion of pre-epiglottic or paraglottic fat (T3)
- Nodes common, often bilateral

### TOP DIFFERENTIALS

- Laryngocele
- Benign mucosal swelling
- Minor salivary tumour

### CHECKLIST

- Vallecula and tongue base; pyriform sinus
- Cartilage and extralaryngeal spread




CT with intravenous contrast enhancement, axial. The lesion extends bilaterally (1), highly indicative of a malignancy. The…



Axial CECT shows similar findings with invasion of bilateral paraglottic fat and left AE fold. Tumor bulges through thyroid cartilage notch. Pyriform sinu…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 526.

LARYNX & TRACHEA · TRAUMA

## Laryngeal Trauma

### IMAGING FINDINGS

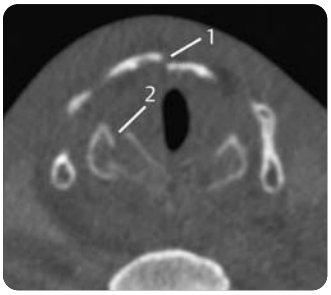
- Thyroid and cricoid cartilage fractures with displacement
- Arytenoid dislocation
- Haematoma and air narrowing the airway

### TOP DIFFERENTIALS

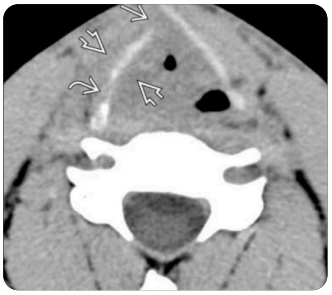
- Cord paralysis
- Unossified cartilage mimicking a fracture

### CHECKLIST

- Secure airway before scanning
- Cricoid fractures and displaced fragments usually need repair




CT, axial bone window. CT at the level of the glottis showed multiple fractures; this figure shows a midline fracture with slight inward…



Axial NECT through the supraglottic larynx shows step-off at the superior thyroid notch. There is a subtle fracture of the right thyroid lamina…

Key facts after Koch, Hamilton, Hudgins and Harnsberger, Diagnostic Imaging: Head and Neck, 3rd ed., p. 278.